Problem 1: Floating Point Increments

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a) Difference between largest and second largest non-infinite number?

largest: 0 1110 111 $= 1.111 \cdot 2^{14-7} = 1.111_2 \cdot 2^7$

s Exp Mantissa

2nd largest: 0 1110 110 $= 1.110 \cdot 2^{14-7} = 1.110_2 \cdot 2^7$

s Exp M

$= 0.001 \cdot 2^7$

$1.00 \cdot 2^4$

$\boxed{1.00_2 \cdot 2^4} = \boxed{16}$

b) Difference between 2nd smallest and smallest positive number

2nd Smallest: 0 . 0000 010 $= 0.010_2 \cdot 2^{1-7} = 0.010 \cdot 2^{-6}$

s exp m

Smallest: 0 0000 001 $= 0.001_2 \cdot 2^{1-7} = 0.001 \cdot 2^{-6}$

s exp m

$= 0.001 \cdot 2^{-6}$

$\boxed{1.00_2 \cdot 2^{-9}} = \boxed{\frac{1}{512}}$